

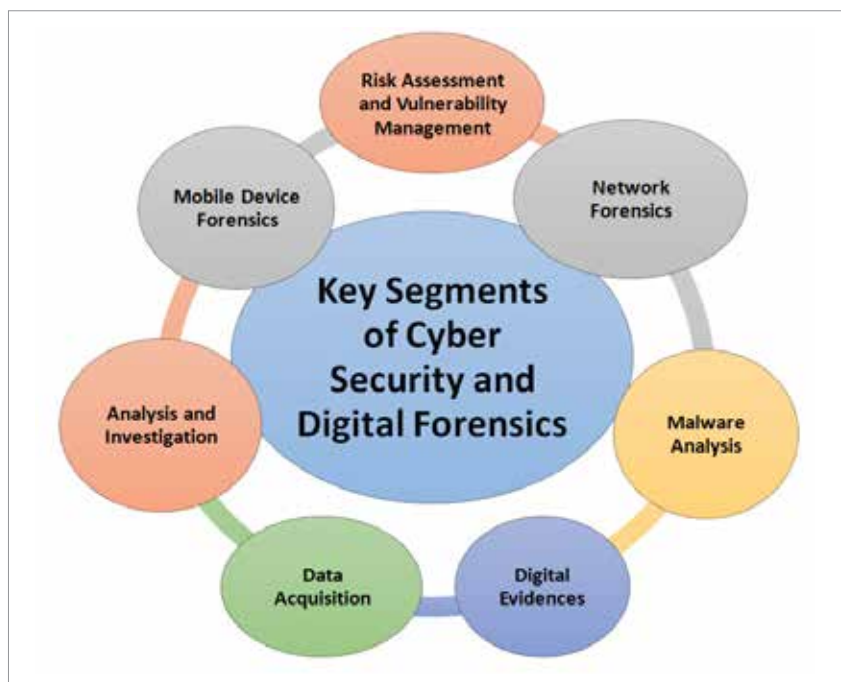


Figure 1: Key segments of cyber security and digital forensics

they target individuals or organisations. The techniques used facilitate account hijacking leading to unauthorised access to online services, financial accounts, or corporate networks.

The rising prevalence of Internet of Things (IoT) devices like smart home assistants, fitness trackers and connected appliances has added additional layers of vulnerability. Since most of these devices are run on a narrow range of security protocols, they become an easy target for a hacker. Compromised IoT devices can serve as a foothold into home or enterprise networks, or be wielded as instruments in large-scale botnet assaults, as in distributed denial-of-service (DDoS) attacks.

Another significant danger is unsecured public networks such as open Wi-Fi hotspots. These may not always employ secure connections or encrypt data, allowing attackers to utilise methods such as packet sniffing, man-in-the-middle (MITM) attacks, or session hijacking to capture sensitive information transmitted over these networks. Even authorised applications

running on users' devices could be security chain bottlenecks if they are over privileged or have any code level vulnerabilities.

In the case of social engineering attacks, psychological manipulation is used to circumvent security measures. Attackers may impersonate as technical support or company staff to gain access to key systems. Supply chain attacks, in which attackers gain access to trusted third-party software or hardware, are a new threat vector that have the potential to take down entire ecosystems of devices.

There are also issues with respect to physical security. If well-designed devices use unencrypted or weak authentication, sensitive data can be leaked in case they are stolen or lost. Portable media devices such as USB drives, external hard drives and smartphones are vulnerable to such risks.

Attackers are now using artificial intelligence (AI) and machine learning (ML) to make their attacks more sophisticated and scalable. For example, AI may be used to produce highly

convincing phishing emails or to automate vulnerability detection.

All these risks require a holistic approach to security including deployment of multi-factor authentication (MFA), use of unique and strong passwords, the update of all software and firmware, encryption of sensitive data, etc. Defences at the network level such as firewalls, intrusion detection systems, and virtual private networks enable secure communication and access. Furthermore, since human error is one of the prime causes behind many security incidents, it is important to build cybersecurity awareness among users.

Free and open source tools for digital forensics and cybersecurity

There is growing interest in free and open source tools for data forensics as they customise and tailor features based on requirements. These tools allow smaller organisations, universities and independent researchers to conduct high-quality forensics analysis without steep licensing fees.

These free and open source tools are used in digital forensics to recover deleted files in storage devices. They help with recovering forensics evidence for computers, mobile devices, and networks that suffered a security breach. They are critical to detecting, analysing and documenting malware attacks, and identifying the origins and methods of cyber intrusions. Information extraction open source tools assist with metadata analysis, mining, sentient analysis, and automatic processing of vast datasets.

Besides, in the open source community, when people collaborate, new ideas and improved software flow continuously. Regular updates, community-contributed add-ons and a plethora of help communities ensure these tools are constantly on the cutting-edge. They are not only used